

Theory of Codes and Cryptography: Practice 1

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1. Provide the different features that the system should meet.

The system should meet the following features:

- **Confidentiality:** only the person with the key can access the information.
- **Identification or authentication of the entity:** making sure the user trying to decrypt the information is the right person.
- **Authorization:** Deciding whether you have the right to access the data and granting the right permissions.
- **Data integrity:** Checking if the data wasn't altered somewhere.
- **Signature:** It links a specific person to a given data ensuring that the person cannot later claim "I didn't send this".

2. Identify and characterize the different entities that interact in the system and describe their main features.

- **User:** person providing data to encrypt or person wanting to decrypt the data.
- **Administrator:** system manager that coordinates access.
- **Symmetric encryption algorithm:** the main algorithm responsible for encrypting and decrypting the data.
- **Public key encryption algorithm:** responsible for encrypting and decrypting the key which the main algorithm uses.
- **Authentication algorithm:** verifies the identity of the user and passes that information to the administrator.

3. Describe the provisional design of the project and explain how the information can be accessed only by the authorized users.

4. Indicate the information that should be additionally stored in order to properly access the original data.

5. Programming language and libraries